



3D AI Career Guidance Mentor — Comprehensive Technical Report

An interactive, real-time 3D WebGL career acceleration platform featuring procedural facial blendshape rigging, 10 full-body skeletal animation clips, ElevenLabs neural voice synthesis, real-time phoneme-to-viseme lip synchronization, ATS resume diagnostics, and deep multi-turn Google Gemini 3.6 Flash reasoning.

Google Gemini 3.6 Flash

Three.js WebGL r128+

ElevenLabs Neural Voice

52 ARKit Blendshapes

Live Code Interview Arena

ATS Score & Skill Diagnostics

8-Stage Career Roadmap

Zero-Leak Server Proxy

1. Executive Summary & Platform Overview

Project Siya solves the fundamental limitations of static, text-only AI career coaching by creating an embodied, highly expressive 3D avatar that interacts simultaneously through **visual, spoken, and computational modalities**. The system merges high-performance graphics rendering with conversational intelligence and specialized technical interview evaluation engines.

CONVERSATIONAL AI ENGINE
Google Gemini 3.6 Flash

3D GRAPHICS & RENDERING
Three.js (60 FPS WebGL)

SPEECH SYNTHESIS (TTS)
ElevenLabs Neural Voice

FACIAL RIGGING STANDARD
52 ARKit Blendshapes

SKELETAL MOTIONS
10 Full-Body Animations

SECURITY MODEL
Server-Side Proxy Isolation

2. System Architecture & High-Level Subsystems

The Siya platform is engineered in modular layers that decouple visual rendering, AI reasoning, speech synthesis, and assessment state management:

3D Avatar & Graphics Engine

Three.js WebGL stage with ACESFilmic tone mapping, 3-point cinematic lighting, Ready Player Me 3D rig with 52 ARKit morph targets, AnimationMixer for 10 skeletal animation clips, and dynamic studio atmosphere themes.

Multi-Turn Conversational Brain

Google Gemini 3.6 Flash with 24-turn conversation memory, automatic model fallback cascade (Gemini 3.6 → 3.5 → 3.1 → LocalBrain), and structured markdown solution formatting with Big-O complexity analysis.

Audio, Speech & Lip Sync

Studio-grade ElevenLabs neural voice synthesis, real-time articulatory phoneme-to-viseme lip sync (11 viseme blends), Web Speech STT, and verbal speech diagnostic (WPM pacing & filler word detection).

AI Mock Interview Arena

Split-screen HackerRank-style live coding assessment studio supporting JavaScript, Python, TypeScript, Java, C++, and Go with automated test case runner, voice hints, and STAR scorecard evaluation.

ATS Resume Intelligence

Multi-format PDF/DOCX resume tokenizer using `pdfjs-dist`, ATS readiness scoring (0-100), 200+ skill

8-Stage Career Discovery Studio

Interactive 8-stage questionnaire assessing engineering archetypes, seniority transitions, market salary

taxonomy detection, and Staff+ architectural skill gap inspection.

benchmarks, compensation targets, and 12-month career progression roadmaps.

3. 3D Avatar, ARKit Blendshapes & Skeletal Motion Matrix

Siya features an authoritative expression state manager (`ExpressionManager`) that lerps facial blendshapes over time and layers natural idle life (breathing, spontaneous blinking bursts, microphone loudness tracking) directly onto the 3D rig:

EXPRESSION / MOTION	TRIGGER TONE	KEY FACIAL BLENDSHAPES	SKELETAL ANIMATION CLIP	BEHAVIOR & ARTICULATION
<code>neutral</code>	Baseline Composure	Smile (0.04), calm brows	<code>Idle</code>	Natural chest breathing, soft random eye blinking
<code>greeting</code>	Welcome & Intro	Smile (0.55), brow raise (0.20), wide eyes	<code>Standing_Greeting</code>	Warm introductory hand-waving gesture
<code>confident</code>	Solutions & Architecture	Smile (0.22), dimples (0.20), steady gaze	<code>Talking_2</code>	Poised, authoritative technical lead posture
<code>thinking</code>	Analysis & Debugging	Brow raise (0.24), furrow (0.20), lip pucker	<code>Talking_0</code>	Contemplative problem-solving posture
<code>happy</code>	Praise & Celebration	Wide smile (0.60), squint (0.18), jaw open	<code>Talking_1</code>	Delighted celebratory gesture
<code>empathetic</code>	Critique & Coaching	Soft smile (0.18), inner brow raise (0.18)	<code>Talking_0</code>	Warm, supportive mentorship stance
<code>surprised</code>	Outages & Warnings	Eye wide (0.70), jaw open (0.28), brow up	<code>Talking_0</code> / <code>Terrified</code>	Startled high-alert reaction
<code>funnyFace</code>	Humor & Jokes	Mouth left, cheek puff, brow raise	<code>Laughing</code>	Cheerful laugh and witty banter posture
<code>Rumba</code>	Offer Victory Party	Delighted smile, energetic expressions	<code>Rumba</code>	Full-body celebratory victory dance

4. Multi-Turn AI Intelligence & Solution Generation

Siya's conversational reasoning engine utilizes **Google Gemini 3.6 Flash** with an expanded token limit (2048 tokens) and structured solution delivery:

- **Full Chat Context Retention:** Maintains up to 24 turns of conversation context in memory, allowing candidates to ask deep follow-up questions ("*explain line 4*", "*how to optimize for space?*").
- **Rich Code & Markdown Output:** Complete implementations in Python, JavaScript, TypeScript, Go, Java, and C++ with full comments, edge-case explanations, and Big-O complexity analysis.
- **Dual-Channel Speech Formatter (`formatForSpeech`):** The chat window renders clean Markdown syntax with one-click copy buttons, while the neural voice speaks a clean spoken summary aloud.

```
// Sample Solution Generation Response Format:
[EXPRESSION: confident]
The optimal approach for Two Sum is a One-Pass Hash Map in O(N) Time and O(N) Space.

def two_sum(nums: list[int], target: int) -> list[int]:
```

```
seen = {}  
for i, num in enumerate(nums):  
    complement = target - num  
    if complement in seen:  
        return [seen[complement], i]  
    seen[num] = i  
return []
```

- * Time Complexity: $O(N)$ – Single pass with $O(1)$ hash map lookups.
- * Space Complexity: $O(N)$ – Stores up to N elements in dictionary.

5. Audio, Neural Speech Synthesis & Lip Synchronization

Siya bridges spoken audio and visual lip articulation using ElevenLabs neural voice synthesis and real-time phoneme-to-viseme crossfading:

**Neural Voice Pipeline**

Streamed via backend proxy to ElevenLabs `eleven_turbo_v2_5` model with natural cadence, pitch variation, and zero robotic artifacts. Native Web Speech API acts as an instant offline fallback.

**Articulatory Lip Sync Engine**

Maps English letters to 11 Ready Player Me visemes (`viseme_aa`, `viseme_E`, `viseme_I`, `viseme_O`, `viseme_U`, `viseme_PP`, `viseme_FF`, `viseme_TH`, `viseme_SS`, `viseme_CH`, `viseme_kk`) with dynamic envelope filtering.

**Speech Pacing Diagnostic**

Monitors verbal answer pacing in real-time, calculating Words Per Minute (WPM) against the optimal interview range (125–165 WPM).

**Filler Word Detection**

Detects and counts vocal filler words ("um", "uh", "like", "basically", "literally", "you know") to help candidates build crisp, confident delivery.

6. Live Mock Interview Arena & Resume Intelligence

Siya provides a comprehensive interview ecosystem combining algorithmic coding challenges with behavioral STAR analysis and resume diagnostics:

- **Split-Screen Code Arena:** Full multi-language editor with line numbering, automatic syntax indentation, and test case verification.
- **Voice Hints & Real-Time Guidance:** Candidates can ask Siya for contextual algorithmic hints without spoiling the entire solution.
- **Targeted JD Simulator:** Extracts key requirements from raw job descriptions to conduct custom role-specific interview rounds.
- **ATS Resume Parser:** Analyzes PDF and Word documents to calculate an ATS compatibility score (0–100), extract categorized skill clouds, and identify Staff-level architectural skill gaps.

7. Security Architecture & Verification

**Zero-Key-Leakage Security Model**

All API keys (`GEMINI_API_KEY`, `ELEVENLABS_API_KEY`) are securely isolated on the backend Node.js Vite proxy middleware (`/api/gemini`, `/api/elevenlabs/tts`). Keys are never bundled into client JavaScript, logged in browser storage, or exposed in frontend network requests.

VERIFICATION TEST	TOOL / COMMAND	RESULT	STATUS
Production Bundle Compilation	<code>npm run build</code> (Vite v5.4.21)	55 modules transformed, zero syntax errors, assets minified	PASSED
Multi-Turn Gemini Integration	Node.js Test Simulation	Context retention across turns, code generation, expression extraction	PASSED
Git Version Control Sync	<code>git push origin main</code>	Clean working tree, commit <code>e59bff5</code> synchronized with remote	PASSED

